

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Federal Energy Regulatory Commission

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Docket No. RM26-4

COMMENTS OF CTC GLOBAL CORPORATION

In response to the Federal Energy Regulatory Commission’s (“Commission” or “FERC”) Advance Notice of Proposed Rulemaking (“ANOPR”) in the above-captioned docket,¹ CTC Global, Inc. (“CTC”)² respectfully submits comments highlighting the role of advanced conductors in the interconnection of large loads.

I. INTRODUCTION

FERC’s ANOPR correctly identifies the challenges posed by the rapid growth of large loads and the urgency of ensuring timely and orderly interconnection. Reconductoring with advanced conductors is a proven, fast, and cost-effective way to create transmission headroom on existing corridors. Carbon composite core advanced conductors can double the capacity on existing structures because they do not have the increase in weight and sag necessary when capacity is increased with early and mid-1900s steel core conductors like ACSR and ACSS. Because of this strength and weight advantage, advanced conductors allow for something traditional conductors cannot do: significantly increase capacity through reconductoring. This speed-to-power approach is key when discussing steps that can and must be taken in order to add new large loads to the power system.

In cases where rebuilds and greenfield transmission is needed, carbon composite core advanced conductors —such as the ACCC Conductor— provide both a cost savings due to reduced tower height or fewer towers, elements that also reduce the risk of siting delays. For many large-load and hybrid interconnections, these attributes materially reduce the scope and cost of network upgrades, shorten implementation timelines, and therefore directly advance the ANOPR’s objectives.

Should FERC decide to invoke authority over the interconnection of large loads greater than 20 megawatts (MW), we urge the Commission to require transmission providers to explicitly evaluate reconductoring with commercially available advanced conductors as a standard

¹ANOPR, *Ensuring the Timely and Orderly Interconnection of Large Loads*, FERC Docket No. RM26-4 (Oct 27, 2025).

² CTC Global is the world’s leading manufacturer of carbon core advanced conductors, with over 124,000 miles of its ACCC Conductor in service worldwide. CTC is a US-based company, headquartered in Irvine, California where it manufactures carbon core for ACCC Conductors using US-sourced components. CTC also produces ACCC around the world from four other manufacturing facilities.

alternative in interconnection and planning studies before ordering more extensive network upgrades.

FERC’s Order No. 1920 has already directed transmission providers to consider alternative transmission technologies, including advanced conductors, when evaluating potential long-term regional transmission facilities, and to explain how they will consider such technologies in both long-term and existing regional planning. Separately, Order No. 2023’s interconnection reforms require transmission providers to evaluate alternative transmission technologies, such as advanced conductors, when conducting a cluster study.

II. ADVANCED CONDUCTORS SHOULD BE TREATED AS A STANDARD INTERCONNECTION OPTION

Advanced conductors should be established as a standard interconnection option for large loads because they directly advance the ANOPR’s stated goals of “minimizing the need for costly network upgrades”³ and “ensur[ing] the timely and orderly addition of large loads to the transmission system.”⁴ Reconductoring existing lines with advanced conductors can often double capacity in 6–24 months by reusing towers and rights-of-way, and avoiding many siting reviews.⁵ These outcomes that lower the network upgrades an interconnection study would otherwise assign to a large-load customer.

Further, reconductoring with advanced conductors is the only way to materially increase transmission capacity in a matter of months instead of years. This is critical to connecting existing generation to demand, as well as unleashing the 2.6 Terawatts of generation capacity currently sitting in interconnection queues.⁶ For more information about how advanced conductors can rapidly and affordably increase transmission capacity, see the attached case studies. While 125,000 miles of ACCC have been deployed globally over the past two decades, less than 10% of that has been in the United States. Yet, researchers at the University of California, Berkeley in an independent, peer-reviewed study, found that **98% of power lines in the United States were good candidates for reconductoring** with carbon composite core advanced conductors.⁷

³ ANOPR P. 20.

⁴ *Id.* P. 24.

⁵ PNAS, *Accelerating Transmission Capacity Expansion by Using Advanced Conductors in Existing Right-of-Way* (September 2024), <https://www.pnas.org/doi/10.1073/pnas.2411207121>.

⁶ Will Gorman et al., *Grid Connection Barriers To New-Build Power Plants In the United States*, Energy Markets & Policy, Berkeley Lab (Jan. 13, 2025), <https://emp.lbl.gov/news/grid-connection-barriers-new-build-power-plants-united-states>.

⁷ See fn. 5. That study found in part: It currently takes 10 to 15 years to build a new power line and the US is building transmission lines at a lower rate than it was in the past decade. Without sufficient capacity, energy projects often sit in limbo for years as transmission operators study what upgrades—if any—are needed to accommodate the increased loads. The authors modeled various scenarios to determine if replacing existing transmission conductors with those made with advanced composite-core materials—a process known as reconductoring—could provide a pathway to faster grid expansion. Based on the authors' projections, it is cheaper—and quicker—for utility companies to replace the 53,000 existing transmission lines with advanced composite-core materials than it is to build entirely new transmission lines. They assert that doing so would reduce wholesale electricity costs by 3% to 4% on average—translating to \$85 billion in system cost savings by 2035 and \$180 billion by 2050.

Lastly, reconductoring with advanced conductors costs far less than building new lines—savings that translate into lower electricity bills for American families and reduced energy costs for businesses. For example, Southern California Edison reconductored 137 miles of its 230 kV Big Creek transmission corridor with ACCC Conductor, avoiding a costly rebuild while eliminating sag violations and adding more than 60% additional line capacity. The project cut costs from \$135 million to \$48 million, reduced completion time from 48 to 18 months, improved safety, lowered line losses by 30%, and delivered significant environmental and customer savings.⁸

All of these performance and delivery characteristics squarely support the ANOPR’s priorities of reduced network upgrade burdens and timely interconnection.

III. CONCLUSION

The Commission’s objectives in this ANOPR—minimizing costly network upgrades and ensuring timely, orderly interconnection of large loads—are directly advanced by reconductoring with advanced conductors. CTC respectfully requests that the Commission require transmission providers to include reconductoring with commercially available advanced conductors as a standard alternative in interconnection and study processes for large loads, and to prioritize the speed and cost savings afforded by reconductoring existing structures where possible. CTC Global stands ready to provide technical data, case studies, and implementation details to assist the Commission and transmission providers.

Respectfully submitted,

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⁸ See attached case studies.



Converted **rebuild to reconductor** project
137 mi ACCC® installed
65% savings in up-front project cost

*"We have used ACCC® to **effectively mitigate sag infractions, increase efficiency, capacity, and reliability** of many of our transmission lines. SCE has decreased fire risk and **built a more resilient grid** and **are continuing to deploy ACCC® at very rapid and cost-effective pace.**"*

Mustafa Ali, Chief Engineer



Reconductor to **2X capacity**
27 mi ACCC® installed
\$10M savings in up-front project cost

*"By utilizing new technology, we were able to **minimize the impact on the local community and keep the power flowing.** The increase in capacity will go a long way to accommodate growth in renewable electricity in the area and **support the transition to net zero.**"*

Rob McDonald, Managing Dir of Transmission



Reconductor **linking new renewable generation**
122 mi ACCC® installed
2X line capacity for critical energy backbone

*"We have tested InfoCore many times over the last two years, and we are seeing **real-world results that indicate this solution is quick and reliable.** Now we are looking down the line at what is to come and are **very excited by the planned developments we are seeing at CTC Global.**"*

Elia Group, Expert Primary Systems



Reconductor project **without de-energizing line**
1,440 mi ACCC® installed
\$43M savings in up-front project cost

*"The savings realized by completing the project while **the line remained energized**, have **saved consumers tens of millions of dollars.** We developed a lot of ACCC® testing with Kinetrics and I tell people that **ACCC® is probably the most tested conductor that I've ever been involved with.**"*

James Berger, (former) Director of Transmission



Reconductor project
94 mi ACCC® installed
2x line capacity to accommodate data centers

*"We have been **using ACCC® in Northern Nevada** as we've **gotten data centers onto our system** as new line builds so that we can **get higher capacity out of our lines.** Our **internal crews prefer working with ACCC®** to a bundled configuration and so we've standardized ACCC® and love using it."*

Devon Kendall, (former) Manager of Civil & Transmission Engineering

Other Recent Projects



Reconductor project
91 mi ACCC® installed



Green Field project
47 mi ACCC® installed



Reconductor project
34 mi ACCC® installed



American Electric Power

AEP utilized 1,440 conductor miles of ACCC Conductor to enhance the capacity of a 345kV double circuit line spanning 120 miles. The upgrade to support load growth boosted the line's capacity from 1,751 amps to 3,099 amps—a **capacity increase of approximately 80%**.

Data Point	ACCC®	ACSR
Project Cost	\$375M	\$418M
Project Speed	33 months	66 months



Southern California Edison

SCE utilized 411 conductor miles of ACCC Conductor to reconstruct the 230kV single circuit line spanning 137 miles of the Big Creek transmission corridor. The upgrade to address conductor sag boosted the line's capacity from 936 amps to 1520 amps— **a capacity increase of approximately 60%**.

Data Point	ACCC®	ACSR
Project Cost	\$48M	\$135M
Project Speed	18 months	48 months

AEP Saved \$43M & Increased Line Capacity



Utility:	American Electric Power
Line configuration:	345kV 120-mile double circuit line
ACCC® install base:	1,440 conductor miles
Project objective:	Needed to increase capacity to accommodate load growth. N-1 line could not take the outage and a rebuild require high capex with 3-4 year timeline

Project details with traditional vs ACCC® Conductor

	ACSR	ACCC®
Rebuild/retrofit required:	Yes	No
De-energizing required:	Yes	No
Conductor type:	ACSR Drake	ACCC® Drake
Project cost:	\$418M	\$375M
Time to completion:	66 months	33 months

ACCC® Solution

1

Increased line capacity faster and at lower cost

Increased the line's capacity by ~80% (1751 amps to 3099 amps) while saving \$43M and 8 months on the project

2

Upgrade completed with uninterrupted service

A temporary single-phase line designed to replace 240 mi while energized (using ACSR would require de-energizing)

3

Reduced line losses

Reduced line loss by 30% enabling energy conservation by freeing up 34 MW of generation capacity

Winner of 2016 Edison Award



These benefits, and the savings realized by completing the project while the line remained energized, have saved consumers tens of millions of dollars.

James Berger, (former) Director of Transmission

Utility:	Southern California Edison
Line configuration:	230 kv 137-mile single circuit line
ACCC [®] conductor installed:	411 conductor miles
Project objective:	SCE needed to rebuild 137 miles of the Big Creek transmission corridor to mitigate sag violations

Project details with traditional vs ACCC[®] conductor

	ACSR	ACCC [®]
Rebuild/retrofit required:	Yes	No
Conductor type:	ACSR Dove	ACCC [®] Dove
Project cost:	\$135M	\$48M
Time to completion:	48 months	18 months

ACCC[®] Solution



1

Increased line capacity

Increased the line's rating from 936 amps to 1520 amps, adding **60%+ more capacity**

2

Sag violation mitigation

Realized **40% improvement in line sag**, mitigating all violations and increasing overall line safety

3

Reduced line losses

Reduced line loss by 30% enabling conservation of generation capacity and saving \$85M in customer costs



Reconductoring with ACCC[®] vs rebuilding saved years of time in permitting and construction, provided significant environmental advantages, and saved tens of millions of dollars in project costs.

SCE

Situation

Need to **add more capacity than in initial project design**



2019: Project design for new line completed

2023: Pre-construction begins post surveys, project planning, land acquisition and permitting

During this time, **generation growth forecasts grew faster than originally planned**

As a result, **original line capacity would be insufficient** to meet the demand

Problem solving with ACCC®

Replace ACSR with ACCC®: **avoid redesign and re-permitting; increase capacity at lower cost**

Complete project with ACSR = **costly redesign and delaying the project** by several years

Utilities replaced ACSR with **ACCC® Conductors** of same weight and diameter and **increased line capacity by 65%**

1,100+ mi of ACCC® to be installed with same line design and right-of-way **meeting construction schedule** for planned 2026 in-service date

Results

\$30M

Cost savings from switching to ACCC® from ACSR

3 Years

Time savings from avoided redesigning, land renegotiations and re-permitting

65%

Extra line capacity added future proofing the grid

~\$17M

Line loss savings lowering total cost of ownership*

Situation

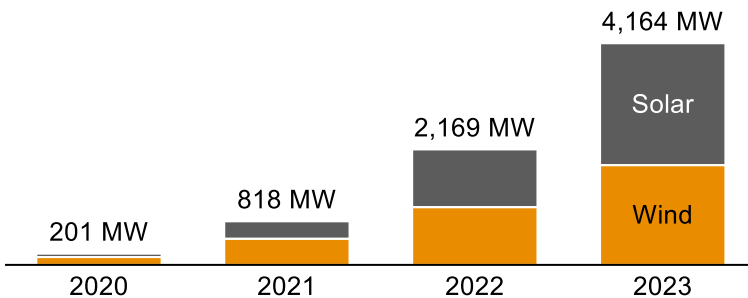


ATCO

New transmission line to address renewable energy growth with **conceptual plans completed in early 2019**

Pre-construction began in Sept 2023 post surveys, project planning, land acquisition and permitting

During this time, **growth in new renewable projects far outpaced original forecasts** and AESO analysts determined that the **original planned capacity of the line would be insufficient** to meet demand



Cumulative New Renewables Capacity in Alberta since 2019

Problem solving with ACCC®

Increasing line capacity using the utilities' **traditional ACSR conductor would have required a costly redesign, and set the project back several years** due to renegotiating land use agreements and re-filing permits

Rather than redesign and face further delays and higher project costs, the AESO utilities (ATCO and AltaLink) **worked with CTC Global to develop a novel solution**

The utilities replaced the traditional ACSR conductor in the original project design with **ACCC® and increased line capacity by 65%**

ACCC® was a "drop-in" replacement with same diameter and weight as the original ACSR conductor. The project will be installing **1,100+ miles of ACCC® with same line design and right-of-way** meeting construction schedule for planned 2026 in-service date

Results

\$30M

Cost savings from switching to ACCC® from ACSR

3 Years

Time savings from avoided redesigning, land renegotiations and re-permitting

65%

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